NGN 2025

SMART HOSPITAL NETWORK

SDN-Based Network Slicing for Ultra-Reliable Low-Latency Healthcare Communication

212010020004 – Ayse Sumeyra Mert

212010020058 – Fatma Zehra Bestil

212010020113 – Betul Korkmaz

KONYA FOOD AND AGRICULTURE UNIVERSITY

KFAU — KGTU

Next-Generation Networks

PRESENTATION OVERVIEW

Agenda

01

Problem Statement

Challenges in conventional hospital networks

02

Proposed Solution

SDN + Network Slicing architecture

03

System Architecture

3-layer framework: Control, Data & Edge

04

Network Slices

URLLC, mIoT, and eMBB slice design

05

KPIs & Evaluation

Performance metrics and targets

06

Risk Management

Technical challenges & mitigation strategies

THE CHALLENGE

Problem Statement

Network Congestion

Critical & non-critical data share same queues – no priority separation

No Prioritization

Static routers/switches treat all packets equally regardless of urgency

Traditional Hospital Network

High Latency

Life-critical signals delayed by bulk transfers (MRI backups, video streaming)

Patient Risk

Even ms-level delays can compromise surgical or ICU monitoring outcomes

"Best-effort" model puts ALL traffic on equal footing – a fatal flaw in critical care.

THE SOLUTION

SDN + Network Slicing

SOFTWARE-DEFINED NETWORKING

- Centralized SDN Controller (ONOS / RYU)
- Decoupled Control & Data Plane
- OpenFlow protocol for flow rule enforcement
- Real-time traffic classification & routing
- Dynamic policy adaptation to network state

+

NETWORK SLICING

- Physical infra → Multiple isolated virtual networks
- Each slice: dedicated QoS & resource guarantees
- Traffic isolation prevents cross-slice interference
- URLLC · mIoT · eMBB slices
- Scalable, future-proof for 5G/6G integration

Together: Programmable, isolated, priority-aware network infrastructure for life-critical healthcare

TRAFFIC ISOLATION

Network Slicing Strategy

SLICE A

HIGHEST PRIORITY

URLLC

Ultra-Reliable Low-Latency

Latency Target:	< 5 ms
Reliability:	99.999%

USE CASES

- Surgical robot control
- ICU patient monitoring
- Emergency alert systems
- Real-time vital signals

SLICE B

MEDIUM PRIORITY

mIoT

Massive Internet of Things

Latency Target:	< 50 ms
Reliability:	99.9%

USE CASES

- Wearable health devices
- Temperature/humidity sensors
- Bed occupancy sensors
- Smart infusion pumps

SLICE C

LOWER PRIORITY

eMBB

Enhanced Mobile Broadband

Latency Target:	< 100 ms
Reliability:	99%

USE CASES

- Staff communications
- Administrative systems
- Guest Wi-Fi access
- Medical imaging upload

3-LAYER FRAMEWORK

System Architecture

LAYER 1 · CONTROL PLANE

SDN Controller (RYU)

Centralized intelligence · Traffic classification · Policy enforcement · OpenFlow rules

LAYER 2 · DATA PLANE

OpenFlow-Enabled Intelligent Switches

Executes controller instructions · Flow table-based forwarding · Deterministic packet delivery

LAYER 3 · EDGE COMPUTING

Distributed Edge Nodes (Local Servers)

Real-time processing near source · Eliminates cloud round-trips · Instant ICU alert generation

FAULT TOLERANCE

Risk Management

RISK 1

Slice Isolation Violation

Lower-priority slice traffic may breach isolation under resource contention, degrading critical URLLC performance

MITIGATION

Strict Resource Reservation

Dedicated bandwidth + CPU capacity permanently allocated to URLLC Slice A via hardware-enforced isolation — non-reallocatable

RISK 2

SDN Controller Single Point of Failure

Centralized SDN controller failure would disrupt global network management, causing total service outage

MITIGATION

Redundant Controller Architecture

Multiple controllers deployed in distributed topology with automatic failover — zero-interruption handoff to backup controller

Both risks mitigated through architectural redundancy and guaranteed resource partitioning.

PERFORMANCE TARGETS

Key Performance Indicators

LATENCY

< 5ms

End-to-End Delay

URLLC slice target for real-time medical data delivery

RELIABILITY

99.999%

Five-Nines Availability

Near-zero packet loss for life-critical data streams

JITTER

Minimal

Delay Variation

Deterministic timing for synchronised medical monitoring

SIMULATION & VALIDATION TOOLS

Mininet

Network emulation platform – virtual topology simulation for SDN slices

Wireshark

Packet-level analysis – latency, jitter & loss measurement across slices

OpenFlow

SDN communication protocol – real-time flow rule enforcement on switches

CONCLUSION

The Future of Smart Hospital Networks

The proposed SDN + Network Slicing architecture transforms passive hospital networks into intelligent, life-critical decision-support infrastructure.

- Mission-critical traffic isolated in dedicated URLLC slices with guaranteed sub-5ms latency
- Centralized SDN control enables real-time programmability and dynamic resource allocation
- Edge computing eliminates cloud dependency for instant patient alert generation
- Modular architecture ready for 5G/6G, AI-based traffic optimization & cloud consensus (Paxos / SMR)

From passive transport layer → Intelligent, adaptive, mission-critical healthcare infrastructure

THANK YOU

NGN PROJECT – SMART HOSPITAL NETWORK

SDN-Based Network Slicing for Ultra-Reliable Low-Latency Healthcare Communication